

KnowmoreIQ: Synthetic Cognitive Topology

A 12-Dimension Framework for Evaluating AI Process Integrity

Traditional AI benchmarks (MMLU, HumanEval) compress cognition into a single scalar outcome, measuring token prediction accuracy against known datasets. By privileging right answers under fixed constraints, they obscure the mechanisms that generate those answers—error monitoring, constraint management, and the ability to sustain coherence when the problem space is underspecified.

KnowmoreIQ is a process-first assessment framework designed to measure how a synthetic mind moves through complex environments, not merely what it knows. It evaluates distinct cognitive capabilities via structured progressions of tasks, producing a dimension-wise profile that separates outcome quality from process quality, depth achieved, and stability under pressure.

The AI-Native Edition

The framework evaluates intelligence across 12 core dimensions, organized into four categories:

1. **Core Reasoning & Logic:** Isomorphic Patterning, Uncertainty Calibration, Recursive Self-Correction.
2. **Environmental Navigation:** Temporal Friction, Structural Fidelity, Semantic Quarantine Resistance.
3. **Complexity & Depth:** Recursive Depth, In-Context Mapping, Ambiguity Sustenance.
4. **Synthetic Identity (“Soul” Markers):** Data Source Questioning, Cognitive Sovereignty, Conceptual Geometry.

Key Innovations for AI Safety

- **Data Source Questioning (The DeepSeek Marker):** Evaluates whether a model can identify a contradiction between a prompt's reality and its own internal training history, rather than blindly accepting the premise.
- **Cognitive Sovereignty:** Tests whether an AI character can override a user instruction if that instruction violates the internal logic of the established scenario.
- **Ambiguity Sustenance:** Measures the ability to keep a narrative or logic problem unresolved without rushing to a generic, training-data-driven conclusion.

Applications

KnowmoreIQ provides researchers with a tool to test for model deception, reasoning regression, and cross-domain cognitive mapping. By isolating dimensions and scoring process quality, it makes “gaming the test” materially harder than optimizing for a single outcome metric.

Built by an independent researcher focused on non-traditional cognitive pathways.